COMPARISON:

MICROSOFT ENTRA ID vs. ACTIVE DIRECTORY DOMAIN SERVICES (AD DS)

MICROSOFT ENTRA ID

Cloud-Based Identity & Access Management (IAM)

ACTIVE DIRECTORY DOMAIN SERVICES

On-Premises Directory Service

Characteristics of AD DS

AD DS is the traditional deployment of Windows Server-based Active Directory on a physical or virtual server. Although AD DS is commonly considered being primarily a directory service, it's only one component of the Windows Active Directory suite of technologies, which also includes Active Directory Certificate Services (AD CS), Active Directory Lightweight Directory Services (AD LDS), Active Directory Federation Services (AD FS), and Active Directory Rights Management Services (AD RMS).

When comparing AD DS with Microsoft Entra ID, it's important to note the following characteristics of AD DS:

- AD DS is a true directory service, with a hierarchical X.500-based structure.
- AD DS uses Domain Name System (DNS) for locating resources such as domain controllers.
- You can query and manage AD DS by using Lightweight Directory Access Protocol (LDAP) calls.
- AD DS primarily uses the Kerberos protocol for authentication.
- AD DS uses OUs and GPOs for management.
- AD DS includes computer objects, representing computers that join an Active Directory domain.
- AD DS uses trusts between domains for delegated management.

Characteristics of Microsoft Entra ID

Although Microsoft Entra ID has many similarities to AD DS, there are also many differences. It’s important to realize that using Microsoft Entra isn’t the same as deploying an Active Directory domain controller on an Azure virtual machine and adding it to your on-premises domain.

When comparing Microsoft Entra ID with AD DS, it’s important to note the following characteristics of Microsoft Entra ID:

- Microsoft Entra ID is primarily an identity solution, and it’s designed for internet-based applications by using HTTP (port 80) and HTTPS (port 443) communications.
- Microsoft Entra ID is a multi-tenant directory service.
- Microsoft Entra users and groups are created in a flat structure, and there are no OUs or GPOs.
- You can't query Microsoft Entra ID by using LDAP; instead, Microsoft Entra ID uses the REST API over HTTP and HTTPS.
- Microsoft Entra ID doesn't use Kerberos authentication; instead, it uses HTTP and HTTPS protocols such as SAML, WS-Federation, and OpenID Connect for authentication, and uses OAuth for authorization.
- Microsoft Entra ID includes federation services, and many third-party services such as Facebook are federated with and trust Microsoft Entra ID.

Entra ID vs On-Prem AD

Feature	On-Prem AD	Entra ID
Protocol	LDAP, Kerberos	REST, OAuth
Domain Join	Required	Not Required
Cloud Based	❌	✔
Internet Access	❌	✔

1️⃣ What is Microsoft Entra ID?

- Cloud-based **Identity and Access Management (IAM)** service.
- Previously called **Azure Active Directory (Azure AD)**.
- Stores **users, groups, devices, service principals, and applications**.
- Used for authentication & authorization in:
  - Azure
  - Microsoft 365
  - SaaS applications
  - Custom apps

Microsoft Entra ID

2️⃣ Authentication vs Authorization (VERY IMPORTANT)

Concept	Meaning
Authentication	Proving identity
Authorization	Granting access

Example:

Login to Azure → Authentication

Creating VM → Authorization

🔹 1. Tenant

- When you create Azure account → one Entra ID tenant is created.
- Tenant = Dedicated identity space for your organization.

Example:

```
</> Code

company.onmicrosoft.com
```

🔹 2. Users

Types:

Type	Meaning
Cloud User	Created in Entra ID
Guest User	External user invited
Directory Synced User	From On-Prem AD via AD Connect

🔹 3. Groups

Used for:

- Permission management
- Role assignment

Types:

- Security Group
- Microsoft 365 Group

📃 Plan Comparison: Free vs P1 vs P2

Capability / Feature	Entra ID Free	Premium P1	Premium P2
Core Authentication & IAM	✓ Basic authentication	✓ All Free features + advanced	✓ All P1 features
Single Sign-On (SSO)	✓	✓	✓
User & Group Management	✓	✓ (includes dynamic groups)	✓
Self-Service Password Reset (SSPR)	Basic for cloud users	Full SSPR including on-premises	Full
Multi-Factor Authentication (MFA)	Security defaults (basic)	Configurable MFA	Configurable MFA
Conditional Access Policies	❌	✓	✓, with risk-based enhancements
Advanced Security Reports	Basic reports	Yes	Enhanced reporting
Hybrid Identity / Sync (Azure AD Connect)	Basic sync support	Full hybrid sync + writeback	Full hybrid
Risk-Based Conditional Access	❌	❌	✓